

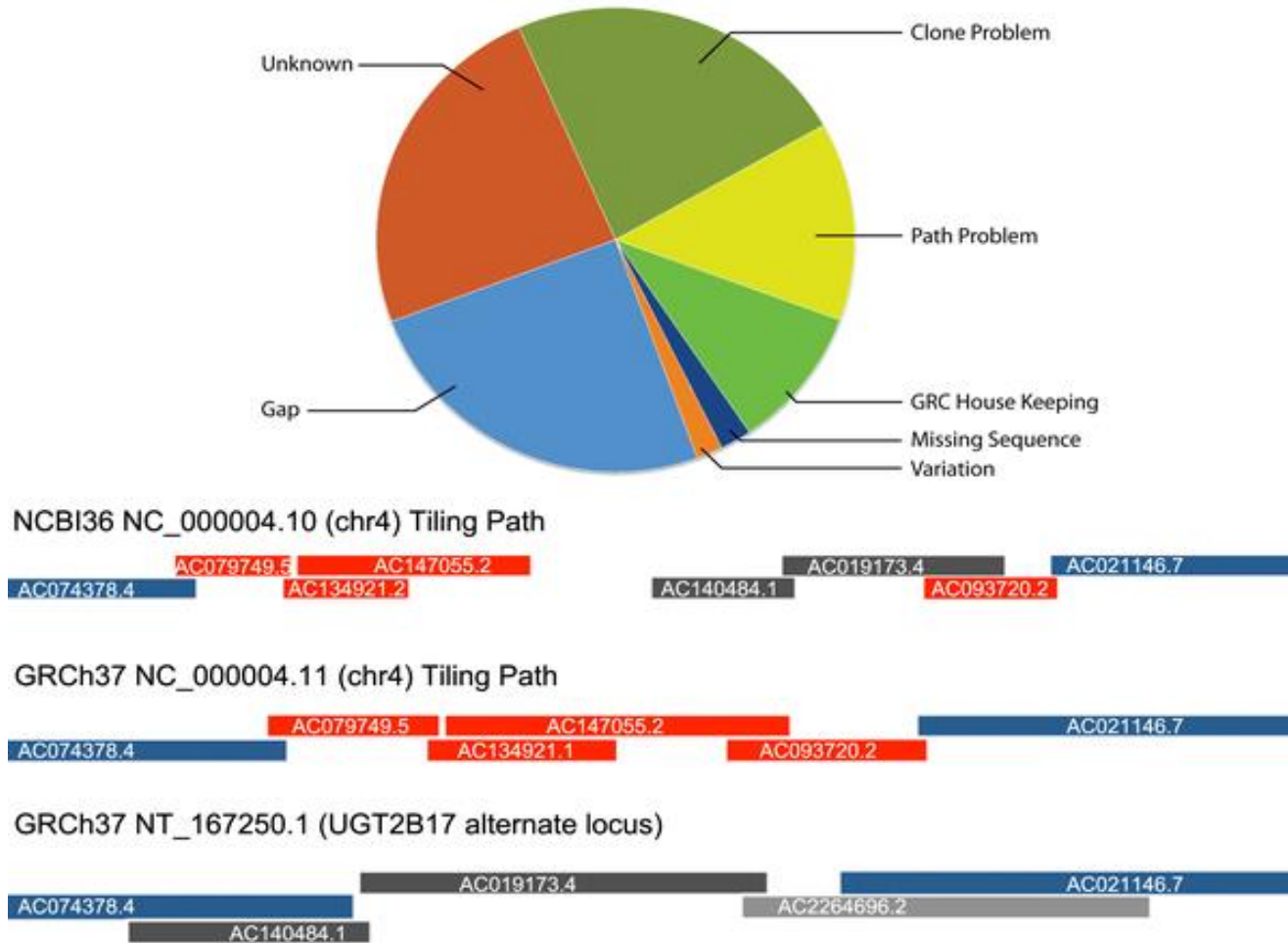
Introduction to Genome Sequences and the UCSC Genome Browser

January 20 2026

Genomic sequence provides a backbone on which to integrate many types of information related to the sequence

- STSs (can be polymorphic or may not be polymorphic)
- Polymorphic markers (RFLPs, microsatellites, SNPs, CNVs)
- Genes (ESTs) and their structure (intron/exons, alternative splicing)
- Regulatory elements
 - Chromatin marks
- Expression Information
- FISH probes
- Chromosome band
- Cross species conservation & homology
- Repetitive regions
-and so much more

No Golden Path!



Church DM, Schneider VA, Graves T, Auger K, et al. (2011) Modernizing Reference Genome Assemblies. PLoS Biol 9(7): e1001091. doi:10.1371/journal.pbio.1001091

<http://www.plosbiology.org/article/info:doi/10.1371/journal.pbio.1001091>

How Do You Interrogate the Genome?

- UCSC Genome Browser (genome.ucsc.edu)
 - University of California Santa Cruz
 - Great for locating things on the genome
- [NCBI](http://www.ncbi.nlm.nih.gov) – National Center for Biotechnology Information (www.ncbi.nlm.nih.gov)
 - NIH
 - Great for getting information about things

Many Other Genome Resources

- ENCODE (Regions important in gene expression)
- ENSEMBL (European Genome Database)
- UniProtKB (Information on proteins)
- OMIM or MalaCards (Disease Associations)
- GTEx (Variants that influence gene expression)
- And so many more
- Genome Browser and NCBI links to many but not all of the resources

Applications for Genome Browser Use

- Linkage/Genome-Wide Association follow up
 - Mendelian Diseases
 - Complex Disorders, QTLs
- Gene Localization for CNVs
- Other Applications
 - Cancer Genetics
 - What gene is disrupted by a breakpoint?
 - RNAseq result interpretation
 - Are functionally related groups also physically related?
 - Future material

Follow up of a linkage region

- How big is the interval?
 - Physical and genetic distances
- What genetic markers are in the interval?
 - Identify gene critical region by genotyping additional markers
- What genes are in the interval?
 - Identify all genes in the gene critical region
- Are the genes in the interval well characterized?
 - Identify best candidate genes
 - Function
 - expression pattern
 - alternative splice forms.

Genome Wide Association Studies (GWAS)

follow up (Future Material)

- Where are **significantly associated** SNPs found in relationship to each other?
- Where are those top SNPS located in relationship to functional regions of the genome i.e. in genes or nearby?
 - **ENCODE** regions (Dr. Landry)
 - **GTEx**
- What other SNPs are in **linkage disequilibrium (LD)** with associated SNPs?

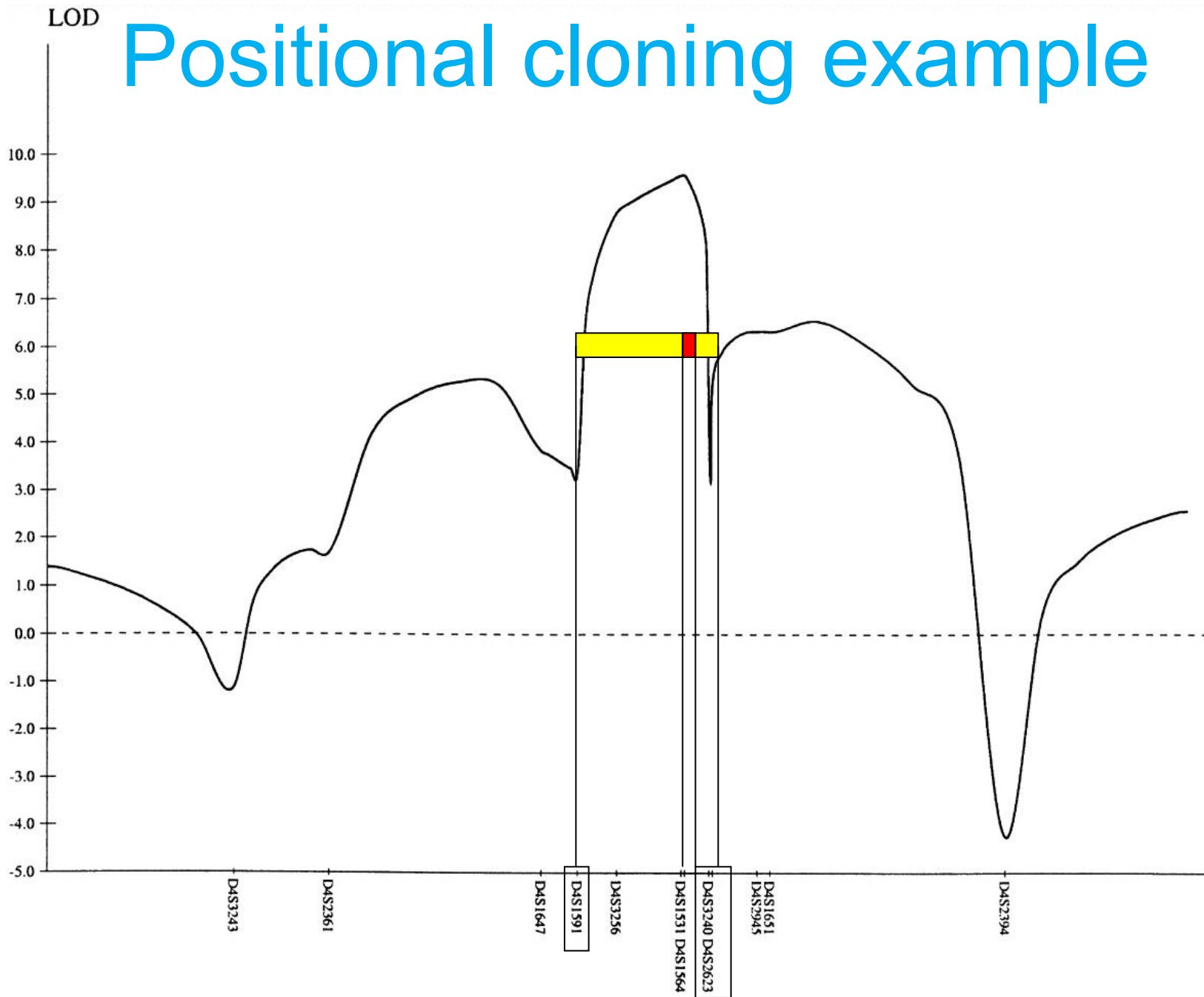
Applications for Genome Information at NCBI

- Once genes of interests are identified NCBI has information on genes and the tools to interrogate the information

Popular NCBI Tools/Databases

- Gene - Integrates many different resources at NCBI
- [Pubmed](#) - Literature searches
- [BLAST/GenBank](#) - Compares a sequence against entries in GenBank
- Variation databases – Find variant information
 - dbVar (CNVs)
 - SNP (rs#)
 - ClinVar
 - [gnomAD](#)
- Expression
 - Multiple source & experiments
 - [GTEx](#)
- NHGRI GWAS Catalog
- OMIM- Online Mendelian Inheritance in Man
 - Summary of disease or gene from the literature

Positional cloning example



Genetic marker terminology for Tutorial

- RFLP- Restriction Fragment Length Polymorphism
- Microsatellite
 - SSLP – Simple Sequence Length Poly
 - STRP- Simple tandem Repeat Poly
 - Dinucleotide repeat
- SNP – Single or Simple Nucleotide Polymorphism
 - Same type of poly as RFLP but not detected by cutting
- D#s and dbSNP 'rs' identifiers